

Hospimed Project Documentation

1. Project Overview

Hospimed is a Flutter-based mobile and web application designed to manage medical/hospital equipment inventory, sales, and analytics. It seamlessly integrates with a Supabase (PostgreSQL) backend, serving as a comprehensive dashboard that handles data potentially synchronized from accounting software (like Tally).

2. Technology Stack

- **Frontend:** Flutter (Dart)
- **Backend & Database:** Supabase (PostgreSQL)
- **Data Visualization:** `fl_chart` package in Flutter
- **State Management:** Provider (e.g., `CompanyProvider`)

3. Core Modules

3.1 Company Management

Allows switching between different companies or branches. Data is filtered on the frontend and backend using `company_name` or `guid`.

3.2 Sales & Invoices

Manages the core sales workflow. - **Sales Invoices:** Tracks invoice numbers, dates, customers, total amounts, discounts, and GST. - **Invoice Items:** Detailed line items for each invoice tracking product names, quantities, unit prices, and GST rates.

3.3 Inventory Management

Tracks medical stock items, including standard cost, selling price, current quantities, categories, and HSN codes.

3.4 Customers & Outstanding Balances

Tracks customer details and their financial standing, including outstanding receivables and payables. Helps in determining overdue amounts and credit limits.

4. Analytics & Reports Engine

Because mobile devices have limited processing power, all heavy calculations for analytics are performed on the Supabase PostgreSQL server using optimized SQL Remote Procedure Calls (RPCs).

4.1 Fast Moving Items

Identifies the most popular items based on total quantity sold over a configurable period (default 30 days). - **Backend:** `get_fast_moving_items` RPC.

4.2 Slow Moving Items

Identifies stock items that are sitting on shelves with little to no sales. - **Backend:** `get_slow_moving_items` RPC.

4.3 Unused Ledgers (Dormant Customers)

Identifies customers who have not made a purchase within the last 6 months (default 180 days). -

Backend: get_unused_ledgers RPC.

4.4 Unused Items (Dead Stock)

Identifies inventory items that have zero sales over the last 6 months but are still in stock. -

Backend: get_unused_items RPC.

4.5 Daily Sales & Estimated Profit

Calculates daily revenue and estimated gross profit. - **Revenue:** Sum of invoice totals for the day. -

Profit: (Selling Price - Standard Cost) * Quantity. Uses a fallback logic where it checks StandardCost, then ItemRate, or defaults to 0 if neither is available. - **Backend:** get_daily_profit RPC.

5. Database Schema Summary

Key tables involved in the operations include: - tally_companies: Base company details. - customers: Customer directory and balances. - stock_items: Inventory and pricing. - sales_invoices & invoice_items: Transaction records. - outstanding_receivables / outstanding_payables: Accounts tracking. - tally_daybook: Daily voucher tracking.

6. Architecture & Security

- **Row Level Security (RLS):** Can be applied on Supabase to ensure users only access their authorized company data.
- **Data Sync:** The database structure includes guid, master_id, and synced_to_tally fields to facilitate two-way or one-way syncing with external ERP systems like Tally.